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Assignment 4

CS 420

Part 1: Analyze the Current Design

1. Identify the primary key of the table.

SessionID is the primary key.

2. Write at least 5 functional dependencies.

FD1: SessionID -> SessionDate, SessionTime, SessionType, HourlyRate, DurationMinutes, StudentID, TutorID, CourseID, RoomID

FD2: StudentID -> StudentName, StudentEmail, Major

FD3: TutorID -> TutorName, TutorEmail

FD4: CourseID -> CourseTitle, Department

FD5: RoomID -> RoomBuilding, RoomCapacity

3. Describe the following anomalies:
Insertion anomaly
If the tutoring center wants to add a new student who has not attended a tutoring session yet, the current table does not have a good place to store that student without also creating a fake session.

Update anomaly
If a student changes their email, the email may need to be updated in many rows because the student can attend multiple sessions. If one row is missed, the database will have inconsistent student email information.

Deletion anomaly
If a session is deleted, important information about a student, tutor, course, or room could also be lost if that was the only row where that information appeared.

Part 2: Normalize to 3NF

STUDENT

STUDENT(StudentID, StudentName, StudentEmail, Major)
PK: StudentID
FK: None

TUTOR

TUTOR(TutorID, TutorName, TutorEmail)
PK: TutorID
FK: None

COURSE

COURSE(CourseID, CourseTitle, Department)
PK: CourseID
FK: None

ROOM

ROOM(RoomID, RoomBuilding, RoomCapacity)
PK: RoomID
FK: None

SESSION

SESSION(SessionID, SessionDate, SessionTime, StudentID, TutorID, CourseID, RoomID, SessionType, HourlyRate, DurationMinute)
PK: SessionID
FK:
StudentID references STUDENT(StudentID)
TutorID references TUTOR(TutorID)
CourseID references COURSE(CourseID)
RoomID references ROOM(RoomID)

Part 3 - 5: Implement in MySQL

In Assignment4.sql file.

Instance Examples

STUDENT

StudentID	StudentName	StudentEmail	Major
1001	Emma Johnson	emma.john@uni.edu	Computer Science
1002	Liam Brown	liam.brown@uni.edu	Information Technology
1003	Sophia Davis	sophia.david@uni.edu	Biology

TUTOR

TutorID	TutorName	TutorEmail
2001	Alice Chen	alice.chen@uni.edu
2002	Bob Martinez	bob.martinez@uni.edu
2003	Carol Smith	carol.smith@uni.edu

COURSE

CourseID	CourseTitle	Department
CS101	Introduction to Programming	Computer Science
CS201	Database Systems	Computer Science
MATH120	College Algebra	Mathematics

ROOM

RoomID	RoomBuilding	RoomCapacity
R101	Science Hall	25
R102	Science Hall	30
R201	Library	12

SESSION

SessionID	SessionDate	SessionTime	StudentID	TutorID	CourseID
1	2026-05-04	09:00:00	1001	2001	CS101
2	2026-05-04	10:30:00	1002	2003	CS201
3	2026-05-05	13:00:00	1003	2005	BIO110
RoomID	SessionType	HourlyRate	DurationMinutes		
R101	One-on-One	20.00	60		
R102	Group	18.00	90		
R201	One-on-One	22.00	60		